

EXP 5: WRITE A PROGRAM FOR FINDING THE BICONNECTED COMPONENTS IN A GIVEN GRAPH.

AIM:

To implement a C program to find the Biconnected Components of a graph represented using an Adjacency Matrix.

PROCEDURE:

1. Start the program.
2. Read the number of vertices.
3. Enter the names of the vertices.
4. Read the **adjacency matrix** of the graph.
5. Check each pair of vertices in the upper triangular part of the matrix.
6. If `graph[i][j] == 1`, print the corresponding pair as a biconnected component.
7. Display all the biconnected components.
8. Stop the program.

PROGRAM:

```
#include <stdio.h>

#define MAX 10

int graph[MAX][MAX];
char vertex[MAX];
int n;

void findBCC()
{
    printf("\nBiconnected Components are:\n");

    for (int i = 0; i < n; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (graph[i][j] == 1)
            {
                printf("{%c, %c}\n", vertex[i], vertex[j]);
            }
        }
    }
}

int main()
{
```

```

printf("Enter number of vertices: ");
scanf("%d", &n);

printf("Enter vertex names:\n");
for (int i = 0; i < n; i++)
{
    scanf(" %c", &vertex[i]);
}

printf("Enter adjacency matrix:\n");
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        scanf("%d", &graph[i][j]);
    }
}

findBCC();

return 0;
}

```

OUTPUT:

D:\>gcc BCC.c -o BCC.exe

D:\>BCC.exe

Enter number of vertices: 4

Enter vertex names:

A

B

C

D

Enter adjacency matrix:

0 1 1 0

1 0 0 0

1 0 0 1

0 0 1 0

Biconnected Components are:

{A, B}

{A, C}

{C, D}

RESULT:

Thus, the biconnected components for the given graph were identified successfully.